Lecture 3
Determinants

3.1 The Determinant of a Matrix
3.2 Evaluation of a Determinant using Elementary Operations
3.3 Properties of Determinants
3.4 Introduction to Eigenvalues
3.5 Application of Determinants

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3.1 The Determinant of a Matrix

- the determinant of a 2 × 2 matrix:

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

$$\Rightarrow \det(A) = |A| = a_{11}a_{22} - a_{21}a_{12}$$

- Note:

$$\left\| \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \right\| = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}$$

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- Ex. 1: (The determinant of a matrix of order 2)

$$\begin{vmatrix} 2 & -3 \\ 1 & 2 \end{vmatrix} = 2(2) - 1(-3) = 4 + 3 = 7$$

$$\begin{vmatrix} 2 & 1 \\ 4 & 2 \end{vmatrix} = 2(2) - 4(1) = 4 - 4 = 0$$

$$\begin{vmatrix} 0 & 3 \\ 2 & 4 \end{vmatrix} = 0(4) - 2(3) = 0 - 6 = -6$$

- Note: The determinant of a matrix can be positive, zero, or negative.

Determinant of any square matrix $A \in M_{nxn}$: is a number will be defined inductively. Defined already for 2×2 matrix. Assume it is defined for any matrix of size $(n-1) \times (n-1)$.

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1(j-1)} & a_{1j} & a_{1(j+1)} & \cdots & a_{1n} \\ \vdots & & & \vdots & \vdots & \vdots & & \vdots \\ a_{(i-1)1} & & \cdots & a_{(i-1)(j-1)} & a_{(i-1)j} & a_{(i-1)(j+1)} & \cdots & a_{(i-1)n} \\ a_{i1} & & \cdots & a_{i(j-1)} & a_{ij} & a_{i(j+1)} & \cdots & a_{in} \\ a_{(i+1)1} & & \cdots & a_{(i+1)(j-1)} & a_{(i+1)j} & a_{(i+1)(j+1)} & \cdots & a_{(i+1)n} \\ \vdots & & & \vdots & \vdots & \vdots & & \vdots \\ a_{n1} & & \cdots & a_{n(j-1)} & a_{nj} & a_{n(j+1)} & \cdots & a_{nn} \end{bmatrix}$$

Choose the entry a_{ij} and delete entirely the ith row and the jth column

There left a matrix of size $(n-1) \times (n-1)$.

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1(j-1)} & a_{1(j+1)} & \cdots & a_{1n} \\ \vdots & & & \vdots & \vdots & & \vdots \\ a_{(i-1)1} & & \cdots & a_{(i-1)(j-1)} & a_{(i-1)(j+1)} & \cdots & a_{(i-1)n} \\ a_{(i+1)1} & & \cdots & a_{(i+1)(j-1)} & a_{(i+1)(j+1)} & \cdots & a_{(i+1)n} \\ \vdots & & & \vdots & \vdots & & \vdots \\ a_{n1} & & \cdots & a_{n(j-1)} & a_{n(j+1)} & \cdots & a_{nn} \end{bmatrix}$$

The *i*th row and *j*th column of A has been deleted.

- Minor of the entry a_{ij} :

The determinant of the matrix determined by deleting the *i*th row and *j*th column of A

$$M_{ij} = \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1(j-1)} & a_{1(j+1)} & \cdots & a_{1n} \\ \vdots & & & \vdots & \vdots & & \\ a_{(i-1)1} & & \cdots & a_{(i-1)(j-1)} & a_{(i-1)(j+1)} & \cdots & a_{(i-1)n} \\ a_{(i+1)1} & & \cdots & a_{(i+1)(j-1)} & a_{(i+1)(j+1)} & \cdots & a_{(i+1)n} \\ \vdots & & & \vdots & \vdots & & \vdots \\ a_{n1} & & \cdots & a_{n(j-1)} & a_{n(j+1)} & \cdots & a_{nn} \end{vmatrix}$$

- Cofactor of a_{ij} :

$$C_{ij} = (-1)^{i+j} M_{ij}$$

- Ex:

$$A=\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$\Rightarrow M_{21}=\begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix} \qquad M_{22}=\begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix}$$

$$\Rightarrow C_{21}=(-1)^{2+1}M_{21}=-M_{21} \qquad C_{22}=(-1)^{2+2}M_{22}=M_{22}$$

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- Notes: Sign pattern for cofactors

$$\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix} \qquad \begin{bmatrix} + & - & + & - \\ - & + & - & + \\ + & - & + & - \\ - & + & - & + \end{bmatrix} \qquad \begin{bmatrix} + & - & + & - & + & \cdots \\ - & + & - & + & - & \cdots \\ + & - & + & - & + & \cdots \\ - & + & - & + & - & \cdots \\ + & - & + & - & + & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \end{bmatrix}$$

3×3 matrix $\quad 4 \times 4$ matrix $\quad n \times n$ matrix

- Notes:

  Odd positions (where $i+j$ is odd) have negative signs, and even positions (where $i+j$ is even) have positive signs.

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- Ex 2: Find all the minors and cofactors of A.

$$A=\begin{bmatrix}0 & 2 & 1\\ 3 & -1 & 2\\ 4 & 0 & 1\end{bmatrix}$$

Sol: (1) All the minors of A.

$$\Rightarrow M_{11}=\begin{vmatrix}-1 & 2\\ 0 & 1\end{vmatrix}=-1,\ M_{12}=\begin{vmatrix}3 & 2\\ 4 & 1\end{vmatrix}=-5,\ M_{13}=\begin{vmatrix}3 & -1\\ 4 & 0\end{vmatrix}=4$$

$$M_{21}=\begin{vmatrix}2 & 1\\ 0 & 1\end{vmatrix}=2,\quad M_{22}=\begin{vmatrix}0 & 1\\ 4 & 1\end{vmatrix}=-4,\ M_{23}=\begin{vmatrix}0 & 2\\ 4 & 1\end{vmatrix}=-8$$

$$M_{31}=\begin{vmatrix}2 & 1\\ -1 & 2\end{vmatrix}=5,\quad M_{32}=\begin{vmatrix}0 & 1\\ 3 & 2\end{vmatrix}=-3,\ M_{33}=\begin{vmatrix}0 & 2\\ 3 & -1\end{vmatrix}=-6$$

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Sol: (2) All the cofactors of A.

$$\because\quad C_{ij}=(-1)^{i+j}M_{ij}$$

$$\Rightarrow C_{11}=+\begin{vmatrix}-1 & 2\\ 0 & 1\end{vmatrix}=-1,\ C_{12}=-\begin{vmatrix}3 & 2\\ 4 & 1\end{vmatrix}=5,\quad C_{13}=+\begin{vmatrix}3 & -1\\ 4 & 0\end{vmatrix}=4$$

$$C_{21}=-\begin{vmatrix}2 & 1\\ 0 & 1\end{vmatrix}=-2,\quad C_{22}=+\begin{vmatrix}0 & 1\\ 4 & 1\end{vmatrix}=-4,\ C_{23}=-\begin{vmatrix}0 & 2\\ 4 & 1\end{vmatrix}=8$$

$$C_{31}=+\begin{vmatrix}2 & 1\\ -1 & 2\end{vmatrix}=5,\quad C_{32}=-\begin{vmatrix}0 & 1\\ 3 & 2\end{vmatrix}=3,\quad C_{33}=+\begin{vmatrix}0 & 2\\ 3 & -1\end{vmatrix}=-6$$

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- Thm 3.1: (Expansion by cofactors)

Let A is a square matrix of order n.

Then the determinant of A is given by

$$(a)\quad \det(A) = |A| = \sum_{j=1}^{n} a_{ij}C_{ij} = a_{i1}C_{i1} + a_{i2}C_{i2} + \cdots + a_{in}C_{in}$$

(Cofactor expansion along the i-th row, i=1, 2,…, n)

or

$$(b)\quad \det(A) = |A| = \sum_{i=1}^{n} a_{ij}C_{ij} = a_{1j}C_{1j} + a_{2j}C_{2j} + \cdots + a_{nj}C_{nj}$$

(Cofactor expansion along the j-th column, j=1, 2,…, n)

- Ex: The determinant of a matrix of order 3

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$\begin{aligned}
\Rightarrow \det(A) &= a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13} && \text{Expansion along the 1st row} \\
&= a_{21}C_{21} + a_{22}C_{22} + a_{23}C_{23} && \text{Expansion along the 2nd row} \\
&= a_{31}C_{31} + a_{32}C_{32} + a_{33}C_{33} && \text{Expansion along the 3rd row} \\
&= a_{11}C_{11} + a_{21}C_{21} + a_{31}C_{31} && \text{Expansion along the 1st column} \\
&= a_{12}C_{12} + a_{22}C_{22} + a_{32}C_{32} && \text{Expansion along the 2nd column} \\
&= a_{13}C_{13} + a_{23}C_{23} + a_{33}C_{33} && \text{Expansion along the 3rd column}
\end{aligned}$$

- Observation: $det(A) = det(A^T)$

- Ex 3: The determinant of a matrix of order 3

$$A=\begin{bmatrix}0 & 2 & 1\\ 3 & -1 & 2\\ 4 & 0 & 1\end{bmatrix}$$

$$\overset{\text{Ex2}}{\Rightarrow} C_{11}=-1,\ C_{12}=5,\quad C_{13}=4$$
$$C_{21}=-2,\ C_{22}=-4,\ C_{23}=8$$
$$C_{31}=5,\quad C_{32}=3,\quad C_{33}=-6$$

Sol:

$$\Rightarrow \det(A)=a_{11}C_{11}+a_{12}C_{12}+a_{13}C_{13}=(0)(-1)+(2)(5)+(1)(4)=14$$
$$=a_{21}C_{21}+a_{22}C_{22}+a_{23}C_{23}=(3)(-2)+(-1)(-4)+(2)(8)=14$$
$$=a_{31}C_{31}+a_{32}C_{32}+a_{33}C_{33}=(4)(5)+(0)(3)+(1)(-6)=14$$
$$=a_{11}C_{11}+a_{21}C_{21}+a_{31}C_{31}=(0)(-1)+(3)(-2)+(4)(5)=14$$
$$=a_{12}C_{12}+a_{22}C_{22}+a_{32}C_{32}=(2)(5)+(-1)(-4)+(0)(3)=14$$
$$=a_{13}C_{13}+a_{23}C_{23}+a_{33}C_{33}=(1)(4)+(2)(8)+(1)(-6)=14$$

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- Ex 5: (The determinant of a matrix of order 3)

$$A=\begin{bmatrix}0 & 2 & 1\\ 3 & -1 & 2\\ 4 & 0 & 1\end{bmatrix}\Rightarrow \det(A)=?$$

Sol:

$$C_{11}=(-1)^{1+1}\begin{vmatrix}-1 & 2\\ 0 & 1\end{vmatrix}=-1\qquad C_{12}=(-1)^{1+2}\begin{vmatrix}3 & 2\\ 4 & 1\end{vmatrix}=(-1)(-5)=5$$

$$C_{13}=(-1)^{1+3}\begin{vmatrix}3 & -1\\ 4 & 0\end{vmatrix}=4$$

$$\Rightarrow \det(A)=a_{11}C_{11}+a_{12}C_{12}+a_{13}C_{13}$$
$$=(0)(-1)+(2)(5)+(1)(4)$$
$$=14$$

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- Notes:

  The row (or column) containing the most zeros is the best choice for expansion by cofactors .

- Ex 4: (The determinant of a matrix of order 4)

$$A=\begin{bmatrix} 1 & -2 & 3 & 0 \\ -1 & 1 & 0 & 2 \\ 0 & 2 & 0 & 3 \\ 3 & 4 & 0 & -2 \end{bmatrix} \Rightarrow \det(A)=?$$

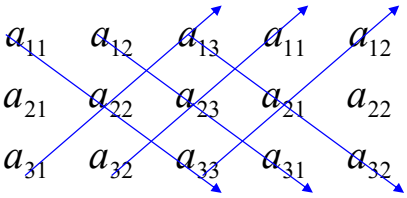
Sol:

$$\begin{aligned}\det(A) &= (3)(C_{13})+(0)(C_{23})+(0)(C_{33})+(0)(C_{43}) \\ &= 3C_{13} \\ &= 3(-1)^{1+3}\begin{vmatrix} -1 & 1 & 2 \\ 0 & 2 & 3 \\ 3 & 4 & -2 \end{vmatrix} \\ &= 3\left[(0)(-1)^{2+1}\begin{vmatrix} 1 & 2 \\ 4 & -2 \end{vmatrix}+(2)(-1)^{2+2}\begin{vmatrix} -1 & 2 \\ 3 & -2 \end{vmatrix}+(3)(-1)^{2+3}\begin{vmatrix} -1 & 1 \\ 3 & 4 \end{vmatrix}\right] \\ &= 3[0+(2)(1)(-4)+(3)(-1)(-7)] \\ &= (3)(13) \\ &= 39\end{aligned}$$

- The determinant of a matrix of order 3:

$$A=\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

Subtract these three products.



Add these three products.

$$\Rightarrow \det(A) = |A| = a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{31}a_{22}a_{13} - a_{32}a_{23}a_{11} - a_{33}a_{21}a_{12}$$

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- Ex 5:

$$A=\begin{bmatrix} 0 & 2 & 1 \\ 3 & -1 & 2 \\ 4 & -4 & 1 \end{bmatrix}$$

$-4 \quad 0 \quad 6$

$0 \quad 16 \quad -12$

$$\Rightarrow \det(A) = |A| = 0 + 16 - 12 - (-4) - 0 - 6 = 2$$

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- Upper triangular matrix:

  All the entries below the main diagonal are zeros.

- Lower triangular matrix:

  All the entries above the main diagonal are zeros.

- Diagonal matrix:

  All the entries above and below the main diagonal are zeros.

- Note:

  A matrix that is both upper and lower triangular is called diagonal.

- Ex:

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a_{22} & a_{23} \\ 0 & 0 & a_{33} \end{bmatrix} \qquad \begin{bmatrix} a_{11} & 0 & 0 \\ a_{21} & a_{22} & 0 \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \qquad \begin{bmatrix} a_{11} & 0 & 0 \\ 0 & a_{22} & 0 \\ 0 & 0 & a_{33} \end{bmatrix}$$

upper triangular lower triangular diagonal

- Thm 3.2: (Determinant of a Triangular Matrix)

  If A is an $n \times n$ triangular matrix (upper triangular, lower triangular, or diagonal), then its determinant is the product of the entries on the main diagonal. That is

$$\det(A) = |A| = a_{11}a_{22}a_{33}\cdots a_{nn}$$

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- Ex 6: Find the determinants of the following triangular matrices.

$$(a)\quad A = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 4 & -2 & 0 & 0 \\ -5 & 6 & 1 & 0 \\ 1 & 5 & 3 & 3 \end{bmatrix} \qquad (b)\quad B = \begin{bmatrix} -1 & 0 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 0 & -2 \end{bmatrix}$$

Sol:

$(a)\quad |A| = (2)(-2)(1)(3) = -12$

$(b)\quad |B| = (-1)(3)(2)(4)(-2) = 48$

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3.2 Evaluation of a determinant using elementary operations

Recall

Let A be a square matrix. Elementary row operations are

- r_{ij} - exchange *i-th* row and *j-th* row of A
- $r_i^{(k)}$ – multiplying the *i-th* row of A by scalar k
- $r_{ij}^{(k)}$ – adding k times of the *i-th* row to the *j-th* row of A

Definition

If I denotes the identity matrix – the matrix with 1's on the main diagonal and 0 elsewhere, we denote

$$R_{ij} = r_{ij}(I)\ ; \quad R_i^{(k)} = r_i^{(k)}(I); \quad R_{ij}^{(k)} = r_{ij}^{(k)}(I)$$

They are called elementary matrices and usually denoted by E.

Homework #1: Prove that

$$R_{ij}.A = r_{ij}(A)\ ; \quad R_i^{(k)}.A = r_i^{(k)}(A); \quad R_{ij}^{(k)}.A = r_{ij}^{(k)}(A)$$

*i*th col. *j*th col.

$$I_n = \begin{pmatrix} 1 & \cdots & 0 & \cdots & 0 & \cdots & 0 \\ \vdots & & \vdots & & \vdots & & \vdots \\ 0 & \cdots & 1 & \cdots & 0 & \cdots & 0 \\ \vdots & & \vdots & \ddots & \vdots & & \vdots \\ 0 & \cdots & 0 & \cdots & 1 & \cdots & 0 \\ \vdots & & \vdots & & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & \cdots & 0 & \cdots & 1 \end{pmatrix}$$

*i*th row

*j*th row

*i*th col. *j*th col.

$$R_{ij} = \begin{pmatrix} 1 & \cdots & 0 & \cdots & 0 & \cdots & 0 \\ \vdots & & \vdots & & \vdots & & \vdots \\ 0 & \cdots & 0 & \cdots & 1 & \cdots & 0 \\ \vdots & & \vdots & & \vdots & & \vdots \\ 0 & \cdots & 1 & \cdots & 0 & \cdots & 0 \\ \vdots & & \vdots & & \vdots & & \vdots \\ 0 & \cdots & 0 & \cdots & 0 & \cdots & 1 \end{pmatrix}$$

*i*th row

*j*th row

$$R_i^{(k)} = \begin{array}{c} \begin{matrix} & & i\text{th col.} & & j\text{th col.} & & \\ \end{matrix} \\ \begin{pmatrix} 1 & \cdots & 0 & \cdots & 0 & \cdots & 0 \\ \vdots & & \vdots & & \vdots & & \vdots \\ 0 & \cdots & k & \cdots & 0 & \cdots & 0 \\ \vdots & & \vdots & & \vdots & & \vdots \\ 0 & \cdots & 0 & \cdots & 1 & \cdots & 0 \\ \vdots & & \vdots & & \vdots & & \vdots \\ 0 & \cdots & 0 & \cdots & 0 & \cdots & 1 \end{pmatrix} \end{array} \begin{matrix} \\ \\ i\text{th row} \\ \\ j\text{th row} \\ \\ \\ \end{matrix}$$

$$R_{ij}^{(k)} = \begin{array}{c} \begin{matrix} & & i\text{th col.} & & j\text{th col.} & & \\ \end{matrix} \\ \begin{pmatrix} 1 & \cdots & 0 & \cdots & 0 & \cdots & 0 \\ \vdots & & \vdots & & \vdots & & \vdots \\ 0 & \cdots & 1 & \cdots & 0 & \cdots & 0 \\ \vdots & & \vdots & & \vdots & & \vdots \\ 0 & \cdots & k & \cdots & 1 & \cdots & 0 \\ \vdots & & \vdots & & \vdots & & \vdots \\ 0 & \cdots & 0 & \cdots & 0 & \cdots & 1 \end{pmatrix} \end{array} \begin{matrix} \\ \\ i\text{th row} \\ \\ j\text{th row} \\ \\ \\ \end{matrix}$$

3.2 Evaluation of a determinant using elementary operations

- Thm 3.3: (Elementary row operations and determinants)

Let A and B be square matrices.

(a) $B = r_{ij}(A) \Rightarrow \det(B) = -\det(A)$ $\quad (\text{i.e.} \left|r_{ij}(A)\right| = -|A|)$

(b) $B = r_i^{(k)}(A) \Rightarrow \det(B) = k\det(A)$ $\quad (\text{i.e.} \left|r_i^{(k)}(A)\right| = k|A|)$

(c) $B = r_{ij}^{(k)}(A) \Rightarrow \det(B) = \det(A)$ $\quad (\text{i.e.} \left|r_{ij}^{(k)}(A)\right| = |A|)$

- Ex:

$$A=\begin{bmatrix}1 & 2 & 3\\ 0 & 1 & 4\\ 1 & 2 & 1\end{bmatrix} \qquad \det(A)=-2$$

$$A_1=\begin{bmatrix}4 & 8 & 12\\ 0 & 1 & 4\\ 1 & 2 & 1\end{bmatrix} \quad A_2=\begin{bmatrix}0 & 1 & 4\\ 1 & 2 & 3\\ 1 & 2 & 1\end{bmatrix} \quad A_3=\begin{bmatrix}1 & 2 & 3\\ -2 & -3 & -2\\ 1 & 2 & 1\end{bmatrix}$$

$$A_1=r_1^{(4)}(A) \Rightarrow \det(A_1)=\det(r_1^{(4)}(A))=4\det(A)=(4)(-2)=-8$$

$$A_2=r_{12}(A) \Rightarrow \det(A_2)=\det(r_{12}(A))=-\det(A)=-(-2)=2$$

$$A_3=r_{12}^{(-2)}(A) \Rightarrow \det(A_3)=\det(r_{12}^{(-2)}(A))=\det(A)=-2$$

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- Summary:

$$\det(r_{ij}(A))=-\det(A) \Rightarrow \det(A)=-\det(r_{ij}(A))$$

$$\det(r_i^{(k)}(A))=k\det(A) \Rightarrow \det(A)=\frac{1}{k}\det(r_i^{(k)}(A)) \quad (P)$$

$$\det(r_{ij}^{(k)}(A))=\det(A) \Rightarrow \det(A)=\det(r_{ij}^{(k)}(A))$$

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Observation:

1. A row-echelon form of a square matrix is always upper triangular.
2. Using elementary row operations we can always transform any square matrix A into the row-echelon form.

Method to compute determinant:

1st step: Using elementary row operations to transform matrix A into the row-echelon form

2nd step: Using above observation and Thm 3.2 to compute determinant of the row-echelon/upper triangular matrix obtained;

3rd step: Using properties (P) of elementary row operations to compute back the determinant *det(A)*.

- Ex 2: (Evaluation a determinant using elementary row operations)

$$A=\begin{bmatrix}2 & -3 & 10\\ 1 & 2 & -2\\ 0 & 1 & -3\end{bmatrix} \quad \Rightarrow \det(A)=?$$

Sol:

$$\det(A)=\begin{vmatrix}2 & -3 & 10\\ 1 & 2 & -2\\ 0 & 1 & -3\end{vmatrix} \overset{r_{12}}{=} -\begin{vmatrix}1 & 2 & -2\\ 2 & -3 & 10\\ 0 & 1 & -3\end{vmatrix}$$

$$\overset{r_{12}^{(-2)}}{=} -\begin{vmatrix}1 & 2 & -2\\ 0 & -7 & 14\\ 0 & 1 & -3\end{vmatrix} \overset{r_2^{(-\frac{1}{7})}}{=} (-1)\left(\frac{1}{-\frac{1}{7}}\right)\begin{vmatrix}1 & 2 & -2\\ 0 & 1 & -2\\ 0 & 1 & -3\end{vmatrix}$$

$$\overset{r_{23}^{(-1)}}{=} 7\begin{vmatrix}1 & 2 & -2\\ 0 & 1 & -2\\ 0 & 0 & -1\end{vmatrix} = (7)(1)(1)(-1) = -7$$

- Notes: If E is an elementary matrix, then

$$|EA| = |E||A|$$

(1) $E = R_{ij} \Rightarrow |E| = |R_{ij}| = -1$

$$\Rightarrow |EA| = |r_{ij}(A)| = -|A| = |R_{ij}||A| = |E||A|$$

(2) $E = R_i^{(k)} \Rightarrow |E| = |R_i^{(k)}| = k$

$$\Rightarrow |EA| = |r_i^{(k)}(A)| = k|A| = |R_i^{(k)}||A| = |E||A|$$

(3) $E = R_{ij}^{(k)} \Rightarrow |E| = |R_{ij}^{(k)}| = 1$

$$\Rightarrow |EA| = |r_{ij}^{(k)}(A)| = 1|A| = |R_{ij}^{(k)}||A| = |E||A|$$

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- Determinants and elementary column operations

Recall: $det(A) = det(A^T)$

We can do similar things to compute *det(A)* by using columns instead of rows.

Recall: Let A be a square matrix. Elementary row operations are

- r_{ij} – exchange *i-th* row and *j-th* row of A
- $r_i^{(k)}$ – multiplying the *i-th* row of A by scalar k
- $r_{ij}^{(k)}$ – adding k times of the *i-th* row to the *j-th* row of A

Define: Elementary columns operations are

- c_{ij} – exchange *i-th* column and *j-th* column of A
- $c_i^{(k)}$ – multiplying the *i-th* column of A by scalar k
- $c_{ij}^{(k)}$ – adding k times of the *i-th* column to the *j-th* column of A

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- Determinants and elementary column operations
- Thm: (Elementary column operations and determinants)

Let A and B be square matrices.

$$(a)\quad B = c_{ij}(A) \Rightarrow \det(B) = -\det(A) \quad (\text{i.e.} \left|c_{ij}(A)\right| = -|A|)$$

$$(b)\quad B = c_i^{(k)}(A) \Rightarrow \det(B) = k\det(A) \quad (\text{i.e.} \left|c_i^{(k)}(A)\right| = k|A|)$$

$$(c)\quad B = c_{ij}^{(k)}(A) \Rightarrow \det(B) = \det(A) \quad (\text{i.e.} \left|c_{ij}^{(k)}(A)\right| = |A|)$$

- Ex:

$$A = \begin{bmatrix} 2 & 1 & -3 \\ 4 & 0 & 1 \\ 0 & 0 & 2 \end{bmatrix} \qquad \det(A) = -8$$

$$A_1 = \begin{bmatrix} 1 & 1 & -3 \\ 2 & 0 & 1 \\ 0 & 0 & 2 \end{bmatrix} \quad A_2 = \begin{bmatrix} 1 & 2 & -3 \\ 0 & 4 & 1 \\ 0 & 0 & 2 \end{bmatrix} \quad A_3 = \begin{bmatrix} 2 & 1 & 0 \\ 4 & 0 & 1 \\ 0 & 0 & 2 \end{bmatrix}$$

$$A_1 = c_1^{(\frac{1}{2})}(A) \Rightarrow \det(A_1) = \det(c_1^{(4)}(A)) = \frac{1}{2}\det(A) = (\frac{1}{2})(-8) = -4$$

$$A_2 = c_{12}(A) \Rightarrow \det(A_2) = \det(c_{12}(A)) = -\det(A) = -(-8) = 8$$

$$A_3 = c_{23}^{(3)}(A) \Rightarrow \det(A_3) = \det(c_{23}^{(3)}(A)) = \det(A) = -8$$

- Thm 3.4: (Conditions that yield a zero determinant)

  If A is a square matrix and any of the following conditions is true, then $\det(A) = 0$.

  (a) An entire row (or an entire column) consists of zeros.

  (b) Two rows (or two columns) are equal.

  (c) One row (or column) is a multiple of another row (or column).

- Ex:

$$\begin{vmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 4 & 5 & 6 \end{vmatrix} = 0 \qquad \begin{vmatrix} 1 & 4 & 0 \\ 2 & 5 & 0 \\ 3 & 6 & 0 \end{vmatrix} = 0 \qquad \begin{vmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 4 & 5 & 6 \end{vmatrix} = 0$$

$$\begin{vmatrix} 1 & 4 & 2 \\ 1 & 5 & 2 \\ 1 & 6 & 2 \end{vmatrix} = 0 \qquad \begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ -2 & -4 & -6 \end{vmatrix} = 0 \qquad \begin{vmatrix} 1 & 8 & 4 \\ 2 & 10 & 5 \\ 3 & 12 & 6 \end{vmatrix} = 0$$

- Note: Compare the complexity of two methods

Order n	Cofactor Expansion		Row Reduction	
	Additions	Multiplications	Additions	Multiplications
3	5	9	5	10
5	119	205	30	45
10	3,628,799	6,235,300	285	339

Combined methods

- Ex 5: (Evaluating a determinant)

$$A=\begin{bmatrix}-3 & 5 & 2\\ 2 & -4 & -1\\ -3 & 0 & 6\end{bmatrix}$$

Sol:

$$\det(A)=\begin{vmatrix}-3 & 5 & 2\\ 2 & -4 & -1\\ -3 & 0 & 6\end{vmatrix}\overset{C_{13}^{(2)}}{=}\begin{vmatrix}-3 & 5 & -4\\ 2 & -4 & 3\\ -3 & 0 & 0\end{vmatrix}$$

$$=(-3)(-1)^{3+1}\begin{vmatrix}5 & -4\\ -4 & 3\end{vmatrix}=(-3)(-1)=3$$

$$\det(A)=\begin{vmatrix}-3 & 5 & 2\\ 2 & -4 & -1\\ -3 & 0 & 6\end{vmatrix}\overset{r_{12}^{(\frac{4}{5})}}{=}\begin{vmatrix}-3 & 5 & 2\\ \frac{-2}{5} & 0 & \frac{3}{5}\\ -3 & 0 & 6\end{vmatrix}$$

$$=(5)(-1)^{1+2}\begin{vmatrix}\frac{-2}{5} & \frac{3}{5}\\ -3 & 6\end{vmatrix}=(-5)(-\frac{3}{5})=3$$

Combined methods

- Ex 6: (Evaluating a determinant)

$$A=\begin{bmatrix} 2 & 0 & 1 & 3 & -2 \\ -2 & 1 & 3 & 2 & -1 \\ 1 & 0 & -1 & 2 & 3 \\ 3 & -1 & 2 & 4 & -3 \\ 1 & 1 & 3 & 2 & 0 \end{bmatrix}$$

Sol:

$$\det(A)=\begin{vmatrix} 2 & 0 & 1 & 3 & -2 \\ -2 & 1 & 3 & 2 & -1 \\ 1 & 0 & -1 & 2 & 3 \\ 3 & -1 & 2 & 4 & -3 \\ 1 & 1 & 3 & 2 & 0 \end{vmatrix} \underset{r_{25}^{(-1)}}{\overset{r_{24}^{(1)}}{=}} \begin{vmatrix} 2 & 0 & 1 & 3 & -2 \\ -2 & 1 & 3 & 2 & -1 \\ 1 & 0 & -1 & 2 & 3 \\ 1 & 0 & 5 & 6 & -4 \\ 3 & 0 & 0 & 0 & 1 \end{vmatrix}$$

$$=(1)(-1)^{2+2}\begin{vmatrix} 2 & 1 & 3 & -2 \\ 1 & -1 & 2 & 3 \\ 1 & 5 & 6 & -4 \\ 3 & 0 & 0 & 1 \end{vmatrix}$$

Combined methods

$$\overset{c_{41}^{(-3)}}{=}\begin{vmatrix} 8 & 1 & 3 & -2 \\ -8 & -1 & 2 & 3 \\ 13 & 5 & 6 & -4 \\ 0 & 0 & 0 & 1 \end{vmatrix}=(1)(-1)^{4+4}\begin{vmatrix} 8 & 1 & 3 \\ -8 & -1 & 2 \\ 13 & 5 & 6 \end{vmatrix}\underset{r_{21}^{(1)}}{=}\begin{vmatrix} 0 & 0 & 5 \\ -8 & -1 & 2 \\ 13 & 5 & 6 \end{vmatrix}$$

$$=5(-1)^{1+3}\begin{vmatrix} -8 & -1 \\ 13 & 5 \end{vmatrix}$$

$$=(5)(-27)$$

$$=-135$$

3.3 Properties of Determinants

- Thm 3.5: (Determinant of a matrix product)

$$\det(AB) = \det(A)\det(B)$$

- Notes:

(1) $\det(EA) = \det(E)\det(A)$

(2) $\det(A+B) \neq \det(A)+\det(B)$

(3) *(This property is true for square matrices A and B of any order)*

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21}+b_{21} & a_{22}+b_{22} & a_{23}+b_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} + \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ b_{21} & b_{22} & b_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

3.3 Properties of Determinants

$$\begin{vmatrix} a_{11} & a_{12}+b_{12} & a_{13} \\ a_{21} & a_{22}+b_{22} & a_{23} \\ a_{31} & a_{32}+b_{32} & a_{33} \end{vmatrix} = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} + \begin{vmatrix} a_{11} & b_{12} & a_{13} \\ a_{21} & b_{22} & a_{23} \\ a_{31} & b_{32} & a_{33} \end{vmatrix}$$

- Ex 1: (to verify the property on determinant of a matrix product)

$$A=\begin{bmatrix}1 & -2 & 2\\ 0 & 3 & 2\\ 1 & 0 & 1\end{bmatrix} \qquad B=\begin{bmatrix}2 & 0 & 1\\ 0 & -1 & -2\\ 3 & 1 & -2\end{bmatrix}$$

Find $|A|$, $|B|$, and $|AB|$

Sol:

$$|A|=\begin{vmatrix}1 & -2 & 2\\ 0 & 3 & 2\\ 1 & 0 & 1\end{vmatrix}=-7 \qquad |B|=\begin{vmatrix}2 & 0 & 1\\ 0 & -1 & -2\\ 3 & 1 & -2\end{vmatrix}=11$$

$$AB=\begin{bmatrix}1 & -2 & 2\\ 0 & 3 & 2\\ 1 & 0 & 1\end{bmatrix}\begin{bmatrix}2 & 0 & 1\\ 0 & -1 & -2\\ 3 & 1 & -2\end{bmatrix}=\begin{bmatrix}8 & 4 & 1\\ 6 & -1 & -10\\ 5 & 1 & -1\end{bmatrix}$$

$$\Rightarrow |AB|=\begin{vmatrix}8 & 4 & 1\\ 6 & -1 & -10\\ 5 & 1 & -1\end{vmatrix}=-77$$

- Check:

$$|AB| = |A|\,|B|$$

- Thm 3.6: (Determinant of a scalar multiple of a matrix)

If A is an $n \times n$ matrix and c is a scalar, then

$$\det(cA) = c^n \det(A)$$

- Ex 2:

$$A = \begin{bmatrix} 10 & -20 & 40 \\ 30 & 0 & 50 \\ -20 & -30 & 10 \end{bmatrix}, \quad \begin{vmatrix} 1 & -2 & 4 \\ 3 & 0 & 5 \\ -2 & -3 & 1 \end{vmatrix} = 5$$

Find $|A|$.

Sol:

$$A = 10\begin{bmatrix} 1 & -2 & 4 \\ 3 & 0 & 5 \\ -2 & -3 & 1 \end{bmatrix} \Rightarrow |A| = 10^3 \begin{vmatrix} 1 & -2 & 4 \\ 3 & 0 & 5 \\ -2 & -3 & 1 \end{vmatrix} = (1000)(5) = 5000$$

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- Thm 3.7: (Determinant of an invertible matrix)

A square matrix A is invertible (nonsingular) if and only if

$$\det(A) \neq 0$$

Why?

- I identity matrix implies $\det(I) = 1$
- A is invertible iff there exist A^{-1} such that $A.\ A^{-1} = I$
- $det(A.\ A^{-1}) = det(A).det(A^{-1}) = det(I) = 1$

USTH

- Thm 3.7: (Determinant of an invertible matrix)

A square matrix A is invertible (nonsingular) if and only if

$$\det(A) \neq 0$$

- Ex 3: (Classifying square matrices as singular or nonsingular)

$$A = \begin{bmatrix} 0 & 2 & -1 \\ 3 & -2 & 1 \\ 3 & 2 & -1 \end{bmatrix} \qquad B = \begin{bmatrix} 0 & 2 & -1 \\ 3 & -2 & 1 \\ 3 & 2 & 1 \end{bmatrix}$$

Sol:

$|A| = 0 \quad \Rightarrow$ A has no inverse (it is singular).

$|B| = -12 \neq 0 \quad \Rightarrow$ B has an inverse (it is nonsingular).

- Thm 3.8: (Determinant of an inverse matrix)

If A is invertible, then $\det(A^{-1}) = \dfrac{1}{\det(A)}$.

- Thm 3.9: (Determinant of a transpose)

If A is a square matrix, then $\det(A^{T}) = \det(A)$.

- Ex 4:

$$A = \begin{bmatrix} 1 & 0 & 3 \\ 0 & -1 & 2 \\ 2 & 1 & 0 \end{bmatrix}$$

(a) $|A^{-1}| = ?$ (b) $|A^{T}| = ?$

Sol:

$$\because \quad |A| = \begin{vmatrix} 1 & 0 & 3 \\ 0 & -1 & 2 \\ 2 & 1 & 0 \end{vmatrix} = 4$$

$$\therefore \quad |A^{-1}| = \frac{1}{|A|} = \frac{1}{4}$$

$$|A^{T}| = |A| = 4$$

- Equivalent conditions for a nonsingular matrix:

If A is an $n \times n$ matrix, then the following statements are equivalent.

(1) A is invertible.

(2) $A\mathbf{x} = \mathbf{b}$ has a unique solution for every $n \times 1$ matrix $\mathbf{b}$.

(3) $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.

(4) $\det(A) \neq 0$

- Ex 5: Which of the following system has a unique solution?

$$(a)\quad \begin{aligned} 2x_2 - x_3 &= -1 \\ 3x_1 - 2x_2 + x_3 &= 4 \\ 3x_1 + 2x_2 - x_3 &= -4 \end{aligned}$$

$$(b)\quad \begin{aligned} 2x_2 - x_3 &= -1 \\ 3x_1 - 2x_2 + x_3 &= 4 \\ 3x_1 + 2x_2 + x_3 &= -4 \end{aligned}$$

Sol:

(a) $A\mathbf{x} = \mathbf{b}$

$\because$ $|A| = 0$

$\therefore$ This system does not have a unique solution.

(b) $B\mathbf{x} = \mathbf{b}$

$\because$ $|B| = -12 \neq 0$

$\therefore$ This system has a unique solution.

3.4 Introduction to Eigenvalues

- Eigenvalue problem:

  If A is an $n \times n$ matrix, do there exist $n \times 1$ nonzero matrices x such that Ax is a scalar multiple of x ?

- Eigenvalue and eigenvector:

  A : an $n \times n$ matrix

  λ : a scalar

  $\boldsymbol{x}$: a $n \times 1$ nonzero column matrix

Eigenvalue

$$Ax = \lambda x$$

Eigenvector

(The fundamental equation for the eigenvalue problem)

- Ex 1: (Verifying eigenvalues and eigenvectors)

$$A=\begin{bmatrix}1 & 4\\ 2 & 3\end{bmatrix} \qquad x_1=\begin{bmatrix}1\\ 1\end{bmatrix} \qquad x_2=\begin{bmatrix}2\\ -1\end{bmatrix}$$

Eigenvalue

$$Ax_1=\begin{bmatrix}1 & 4\\ 2 & 3\end{bmatrix}\begin{bmatrix}1\\ 1\end{bmatrix}=\begin{bmatrix}5\\ 5\end{bmatrix}=5\begin{bmatrix}1\\ 1\end{bmatrix}=5x_1$$

Eigenvector

Eigenvalue

$$Ax_2=\begin{bmatrix}1 & 4\\ 2 & 3\end{bmatrix}\begin{bmatrix}2\\ -1\end{bmatrix}=\begin{bmatrix}-2\\ 1\end{bmatrix}=-1\begin{bmatrix}2\\ -1\end{bmatrix}=(-1)x_2$$

Eigenvector

- Question:

  Given an $n \times n$ matrix A, how can you find the eigenvalues and corresponding eigenvectors?

- Note:

  $Ax=\lambda x \;\Rightarrow\; (\lambda \mathrm{I}-A)x=0$ (homogeneous system)

  If $(\lambda \mathrm{I}-A)x=0$ has nonzero solutions iff $\det(\lambda \mathrm{I}-A)=0$.

- Characteristic equation of an *nxn* matrix *A*:

$$\det(\lambda \mathrm{I}-A)=\left|(\lambda \mathrm{I}-A)\right|=\lambda^n+c_{n-1}\lambda^{n-1}+\cdots+c_1\lambda+c_0=0$$

- Ex 2: (Finding eigenvalues and eigenvectors)

$$A=\begin{bmatrix}1 & 4\\ 2 & 3\end{bmatrix}$$

Sol: Characteristic equation:

$$|\lambda \mathrm{I}-A|=\begin{vmatrix}\lambda-1 & -4\\ -2 & \lambda-3\end{vmatrix}$$

$$=\lambda^2-4\lambda-5=(\lambda-5)(\lambda+1)=0$$

$$\Rightarrow \lambda=5,-1$$

Eigenvalues: $\lambda_1=5, \lambda_2=-1$

$$(1)\lambda_1=5 \qquad \Rightarrow(\lambda_1\mathrm{I}-A)x=\begin{bmatrix}4 & -4\\ -2 & 2\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix}=\begin{bmatrix}0\\ 0\end{bmatrix}$$

$$\begin{bmatrix}x_1\\ x_2\end{bmatrix}=\begin{bmatrix}t\\ t\end{bmatrix}=t\begin{bmatrix}1\\ 1\end{bmatrix},\ t\neq 0$$

$$(2)\lambda_2=-1 \qquad \Rightarrow(\lambda_2\mathrm{I}-A)x=\begin{bmatrix}-2 & -4\\ -2 & -4\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix}=\begin{bmatrix}0\\ 0\end{bmatrix}$$

$$\begin{bmatrix}x_1\\ x_2\end{bmatrix}=\begin{bmatrix}2t\\ -t\end{bmatrix}=t\begin{bmatrix}2\\ -1\end{bmatrix},\ t\neq 0$$

- Ex 3: (Finding eigenvalues and eigenvectors)

$$A=\begin{bmatrix}1 & 2 & -2\\ 1 & 2 & 1\\ -1 & -1 & 0\end{bmatrix}$$

Sol: Characteristic equation:

$$|\lambda \mathrm{I}-A|=\begin{vmatrix}\lambda-1 & -2 & 2\\ -1 & \lambda-2 & -1\\ 1 & 1 & \lambda\end{vmatrix}=(\lambda-1)(\lambda+1)(\lambda-3)=0$$

The eigenvalues : $\lambda_1=1, \lambda_2=-1, \lambda_3=3$

$$\lambda_1=1 \quad \Rightarrow \lambda_1\mathrm{I}-A=\begin{bmatrix}0 & -2 & 2\\ -1 & -1 & -1\\ 1 & 1 & 1\end{bmatrix}\sim\begin{bmatrix}1 & 0 & 2\\ 0 & 1 & -1\\ 0 & 0 & 0\end{bmatrix}$$

$$\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}=\begin{bmatrix}-2t\\ t\\ t\end{bmatrix} \Rightarrow \text{eigenvectors}: \quad t\begin{bmatrix}-2\\ 1\\ 1\end{bmatrix}, t\neq 0$$

$$\lambda_2=-1 \quad \Rightarrow \lambda_2\mathrm{I}-A=\begin{bmatrix}-2 & -2 & 2\\ -1 & -3 & -1\\ 1 & 1 & -1\end{bmatrix}\sim\begin{bmatrix}1 & 0 & -2\\ 0 & 1 & 1\\ 0 & 0 & 0\end{bmatrix}$$

$$\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}=\begin{bmatrix}2t\\ -t\\ t\end{bmatrix} \Rightarrow \text{eigenvectors}: \quad t\begin{bmatrix}2\\ -1\\ 1\end{bmatrix}, t\neq 0$$

$$\lambda_3 = 3 \quad \Rightarrow \lambda_3 \mathrm{I} - A = \begin{bmatrix} 2 & -2 & 2 \\ -1 & 1 & -1 \\ 1 & 1 & 3 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -2t \\ -t \\ t \end{bmatrix} \Rightarrow \text{eigenvectors}: \quad t \begin{bmatrix} -2 \\ -1 \\ 1 \end{bmatrix}, t \neq 0$$

3.5 Applications of Determinants

- Matrix of cofactors of A:

$$[C_{ij}] = \begin{bmatrix} C_{11} & C_{12} & \cdots & C_{1n} \\ C_{21} & C_{22} & \cdots & C_{2n} \\ \vdots & \vdots & & \vdots \\ C_{n1} & C_{n2} & \cdots & C_{nn} \end{bmatrix} \qquad C_{ij} = (-1)^{i+j} M_{ij}$$

- Adjoint matrix of A:

$$adj(A) = [C_{ij}]^T = \begin{bmatrix} C_{11} & C_{21} & \cdots & C_{n1} \\ C_{12} & C_{22} & \cdots & C_{n2} \\ \vdots & \vdots & & \vdots \\ C_{1n} & C_{2n} & \cdots & C_{nn} \end{bmatrix}$$

- Thm 3.10: (The inverse of a matrix given by its adjoint)

  If A is an $n \times n$ invertible matrix, then

$$A^{-1} = \frac{1}{\det(A)} adj(A)$$

- Ex:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$\Rightarrow \det(A) = ad - bc \qquad \Rightarrow A^{-1} = \frac{1}{\det(A)} adj(A)$$

$$adj(A) = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \qquad = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

- Ex 1 & Ex 2:

$$A = \begin{bmatrix} -1 & 3 & 2 \\ 0 & -2 & 1 \\ 1 & 0 & -2 \end{bmatrix}$$

(a) Find the adjoint of A.

(b) Use the adjoint of A to find A^{-1}

Sol: $\because \quad C_{ij} = (-1)^{i+j} M_{ij}$

$$\Rightarrow C_{11} = +\begin{vmatrix} -2 & 1 \\ 0 & -2 \end{vmatrix} = 4, \quad C_{12} = -\begin{vmatrix} 0 & 1 \\ 1 & -2 \end{vmatrix} = 1, \quad C_{13} = +\begin{vmatrix} 0 & -2 \\ 1 & 0 \end{vmatrix} = 2$$

$$C_{21} = -\begin{vmatrix} 3 & 2 \\ 0 & 2 \end{vmatrix} = 6, \qquad C_{22} = +\begin{vmatrix} -1 & 2 \\ 1 & 2 \end{vmatrix} = 0, \quad C_{23} = -\begin{vmatrix} -1 & 3 \\ 1 & 0 \end{vmatrix} = 3$$

$$C_{31} = +\begin{vmatrix} 3 & 2 \\ -2 & 1 \end{vmatrix} = 7, \qquad C_{32} = -\begin{vmatrix} -1 & 2 \\ 0 & 1 \end{vmatrix} = 1, \quad C_{33} = +\begin{vmatrix} -1 & 3 \\ 0 & -2 \end{vmatrix} = 2$$

$\Rightarrow$ cofactor matrix of A $\Rightarrow$ adjoint matrix of A

$$[C_{ij}] = \begin{bmatrix} 4 & 1 & 2 \\ 6 & 0 & 3 \\ 7 & 1 & 2 \end{bmatrix} \qquad adj(A) = [C_{ij}]^T = \begin{bmatrix} 4 & 6 & 7 \\ 1 & 0 & 1 \\ 2 & 3 & 2 \end{bmatrix}$$

$\Rightarrow$ inverse matrix of A

$$A^{-1} = \frac{1}{\det(A)} adj(A) \qquad \because \quad \det(A) = 3$$

$$= \frac{1}{3}\begin{bmatrix} 4 & 6 & 7 \\ 1 & 0 & 1 \\ 2 & 3 & 2 \end{bmatrix} = \begin{bmatrix} \frac{4}{3} & 2 & \frac{7}{3} \\ \frac{1}{3} & 0 & \frac{1}{3} \\ \frac{2}{3} & 1 & \frac{2}{3} \end{bmatrix}$$

- Check: $AA^{-1} = I$

- Thm 3.11: (Cramer's Rule)

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n &= b_n \end{aligned}$$

$A\mathbf{x} = \mathbf{b}$ $\quad A = [a_{ij}]_{n\times n} = [A^{(1)}, A^{(2)}, \cdots, A^{(n)}]$ $\quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$ $\mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$

$$\det(A) = \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix} \neq 0$$

(this system has a unique solution)

$$A_j = \left[A^{(1)}, A^{(2)}, \cdots, A^{(j-1)}, b, A^{(j+1)}, \cdots, A^{(n)}\right]$$

$$= \begin{bmatrix} a_{11} & \cdots & a_{1(j-1)} & b_1 & a_{1(j+1)} & \cdots & a_{1n} \\ a_{21} & \cdots & a_{2(j-1)} & b_2 & a_{2(j+1)} & \cdots & a_{2n} \\ \vdots & & & \ddots & & & \vdots \\ a_{n1} & \cdots & a_{n(j-1)} & b_n & a_{n(j+1)} & \cdots & a_{nn} \end{bmatrix}$$

(i.e. $\det(A_j) = b_1 C_{1j} + b_2 C_{2j} + \cdots + b_n C_{nj}$)

$$\Rightarrow x_j = \frac{\det(A_j)}{\det(A)}, \qquad j = 1, 2, \cdots, n$$

- Pf:

$$A\,\mathbf{x} = \mathbf{b}, \qquad \det(A) \neq 0$$

$$\Rightarrow \mathbf{x} = A^{-1}\mathbf{b} = \frac{1}{\det(A)} adj(A)\mathbf{b}$$

$$= \frac{1}{\det(A)} \begin{bmatrix} C_{11} & C_{21} & \cdots & C_{n1} \\ C_{12} & C_{22} & \cdots & C_{n2} \\ \vdots & \vdots & & \vdots \\ C_{1n} & C_{2n} & \cdots & C_{nn} \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$$

$$= \frac{1}{\det(A)} \begin{bmatrix} b_1 C_{11} & + & b_2 C_{21} & + & \cdots & + & b_n C_{n1} \\ b_1 C_{12} & + & b_2 C_{22} & + & \cdots & + & b_n C_{n2} \\ & & \vdots & & \vdots & & \\ b_1 C_{1n} & + & b_2 C_{2n} & + & \cdots & + & b_n C_{nn} \end{bmatrix}$$

$$\Rightarrow x_j = \frac{1}{\det(A)}(b_1C_{1j} + b_2C_{2j} + \cdots + b_nC_{nj})$$

$$= \frac{\det(A_j)}{\det(A)} \qquad j = 1,2,\cdots,n$$

Since:

$$(b) \quad \det(A) = |A| = \sum_{i=1}^{n} a_{ij}C_{ij} = a_{1j}C_{1j} + a_{2j}C_{2j} + \cdots + a_{nj}C_{nj}$$

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- Ex 4: Use Cramer's rule to solve the system of linear equations.

$$\begin{array}{rcrcrcl} -x & + & 2y & - & 3z & = & 1 \\ 2x & & & + & z & = & 0 \\ 3x & - & 4y & + & 4z & = & 2 \end{array}$$

Sol:

$$\det(A) = \begin{vmatrix} -1 & 2 & -3 \\ 2 & 0 & 1 \\ 3 & -4 & 4 \end{vmatrix} = 10 \qquad \det(A_1) = \begin{vmatrix} 1 & 2 & -3 \\ 0 & 0 & 1 \\ 2 & -4 & 4 \end{vmatrix} = 8$$

$$\det(A_2) = \begin{vmatrix} -1 & 1 & -3 \\ 2 & 0 & 1 \\ 3 & 2 & 4 \end{vmatrix} = -15, \qquad \det(A_3) = \begin{vmatrix} -1 & 2 & 1 \\ 2 & 0 & 0 \\ 3 & -4 & 2 \end{vmatrix} = -16$$

$$x = \frac{\det(A_1)}{\det(A)} = \frac{4}{5} \qquad y = \frac{\det(A_2)}{\det(A)} = \frac{-3}{2} \qquad z = \frac{\det(A_3)}{\det(A)} = \frac{-8}{5}$$

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